



Bhartiya Vidya Bhavan's
Sardar Patel Institute of Technology
(An Autonomous Institute Affiliated to University of Mumbai)
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

Criteria 7: Institutional Values and Best Practices

7.1. Institutional Values and Social Responsibilities

7.1.4: Water conservation facilities available in the Institution:

- 1. Rain water harvesting**
- 2. Borewell /Open well recharge**
- 3. Construction of tanks and bunds**
- 4. Waste water recycling**
- 5. Maintenance of water bodies and distribution system in the campus**

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1	Water Conservation Policy	2

Policy Document for Water conservation facilities available in the Institution

Bhartiya Vidya Bhavan's, Sardar Patel Institute of Technology has had a water conservation policy right from its inception in 2005. As the institute is located in the heart of the city, there is ample supply of Municipal Water. The institute also experiences good amount of rainfall as it is located in a rainfall heavy region. The campus has also a natural gift in the form of a lake. However, the entire Bhavan's campus tries to optimally use the Municipal Water and use ground water for all its other water needs especially gardening.

Apart from this, all the stakeholders of the college are well educated to use water economically and efficiently through the awareness about water conservation spread through notices time to time, display boards at appropriate places at the campus, etc.

The daily requirement of water in the campus is around 49000 litres. The following measures are taken for the conservation of water:

1. RAIN WATER HARVESTING

The institute has implemented strategies to preserve water resources, such as installing rainwater harvesting pits and using drip irrigation for campus vegetation. All roof runoff is channelized through ducts, and several sump systems have been established to store this water. The entire roof of the main building is utilized for rainwater collection, which is then used for groundwater recharge. The campus's extensive greenery also aids in natural groundwater replenishment. These efforts have significantly raised water levels, even in the dry summer months, making the college self-sufficient in water supply, a situation that was not the case before the roof water harvesting system was installed.

2. OPEN WELL RECHARGE

There is no open well at the campus, however an open natural lake is located in the campus is recharged by rain water. The lake also receives water from the various sources of rain water harvesting.

3. CONSTRUCTION OF CONTOUR TRENCH

The small contour bunds have been constructed at many places of the institute to facilitate the percolation of rainwater.

4. USAGE OF WASTE WATER

There is waste water recycling management system at the campus.

5. MAINTENANCE OF WATER BODIES AND DISTRIBUTION SYSTEMS IN THE CAMPUS

5.1 MAINTENANCE OF WATER DISTRIBUTION SYSTEMS IN THE CAMPUS:

The ground water is pumped into storage tanks located at the rooftop of the institute building. The pumping system is also operational using the Solar based Power Supply. There are four numbers of overhead storage tanks each of 10,000 litres each and one Elevated Service Reservoir in the campus. The water is distributed through well laid pipe network. The drinking water is supplied through a separate set of distribution pipes treated using RO powdered water coolers. The water for all other purpose is supplied through another set of distribution pipes. Water facilities are availed in the entire college without any interruption.

Mr. Hemant Murkutkar, the Maintenance Superintendent, closely supervises the entire distribution system to prevent leakages and wastage of water through joints and valves. Additionally, low-pressure flushes are used to reduce water wastage. The RO watercoolers plants are periodically cleaned and maintained by assigning a specific technician in the maintenance department. The institute has the cleaning schedule of all overhead water storage tanks once in six months.



5.2 MAINTENANCE OF WATER BODIES AND DISTRIBUTION SYSTEMS IN THE CAMPUS:

□ Within the campus of the institute has one lake which is closely located to the Bhavan's Nature and Adventure Centre. The staff at this centre maintain the lake regularly.

□ Water bodies are being used for fish cultivation too.

□ Rain water from different buildings of the campus are taken to the lake in such a way that they are not contaminated.

□ Rooftops of the institute building are also kept cleaned under Mr. Hemant Murkutkar, Maintenance Superintendent so that unnecessary garbage does not make the rain water dirty.

□ Water reservoirs are kept above the roof top and also there are underground reservoirs.

□ They are directly connected with the water supply of Municipal Corporation, purified water come to the roof top tanks by pumping the water from underground reservoir.

□ The security guards of the institute strictly monitor the water pumping system so that no water overflow occurs.

□ From the roof top tanks water is distributed through PVC pipe lines.

□ The water from the open natural lake is used for watering the plants. During rainy season, the rain water is also used for gardening.

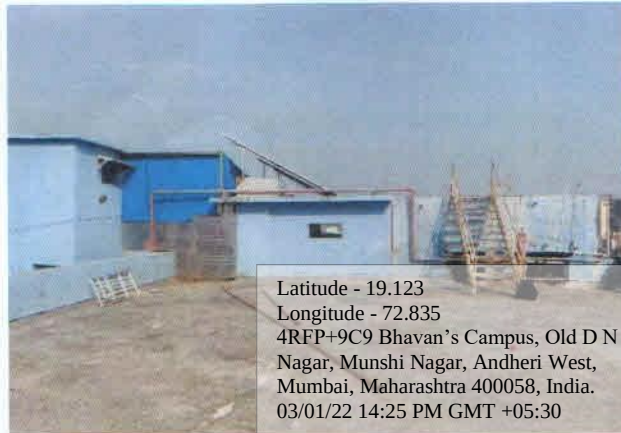
Geo-tagged photographs of the facilities along with Displays:

Water Tank and Distribution System Ground Floor with Solar based Power Supply



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Water Supply Tanks on roof top of the institute building

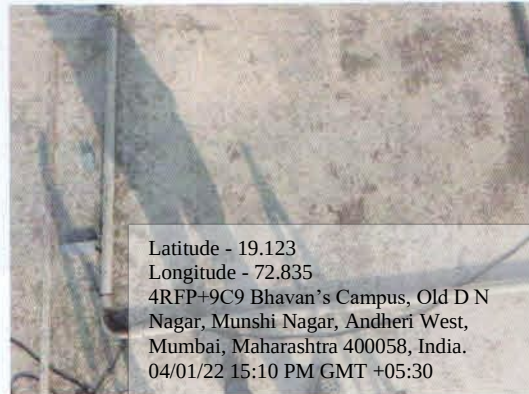


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Water Distribution Piping System

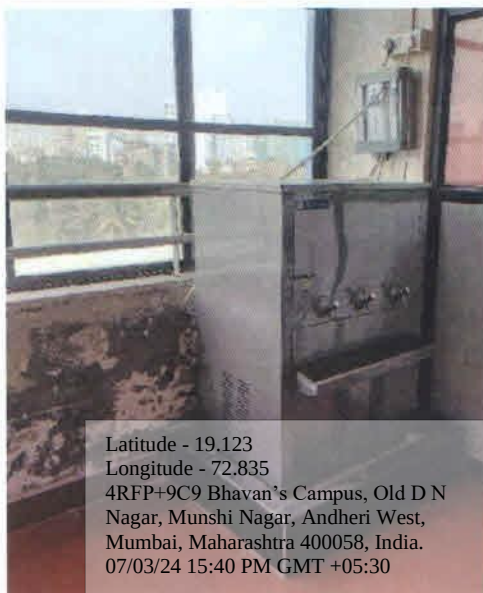


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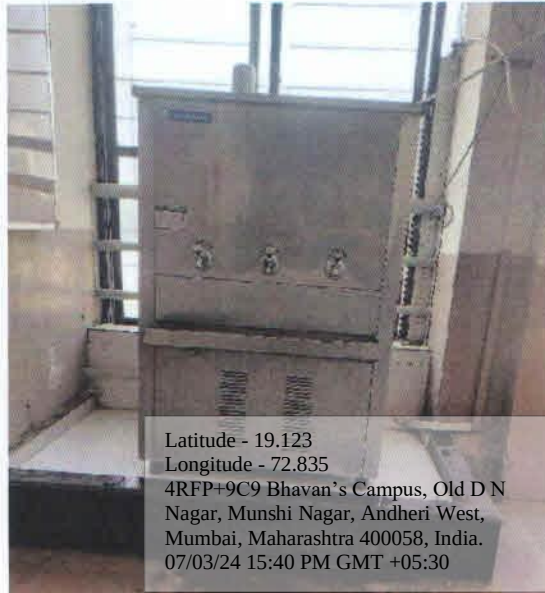


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Water coolers for drinking water

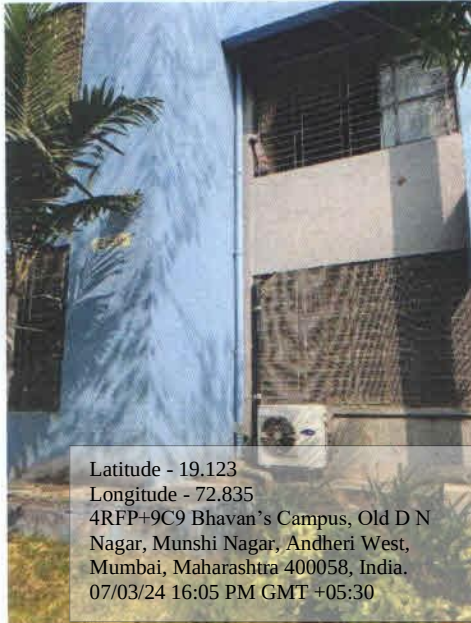


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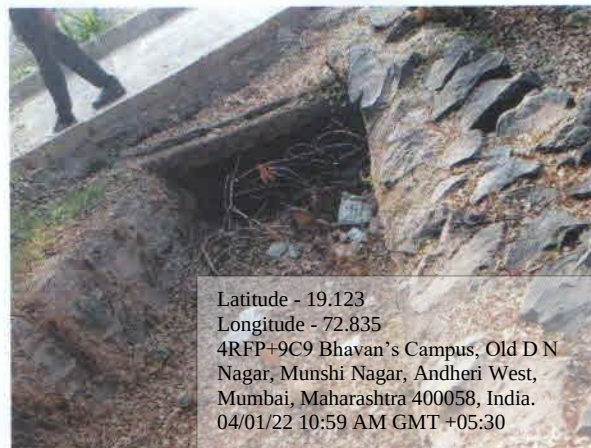


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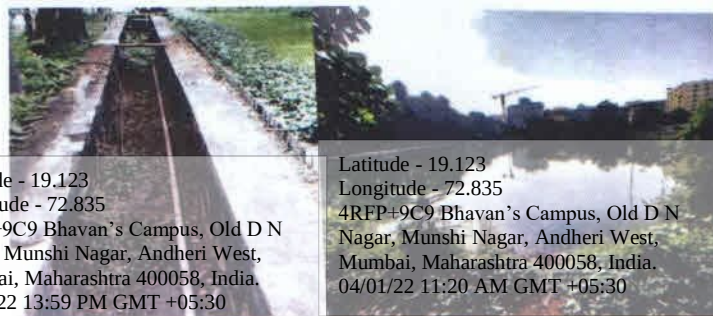
Main rainwater pipe progressing from rooftop



Stormwater Drain



Stormwater drain with pipe carrying water for gardening



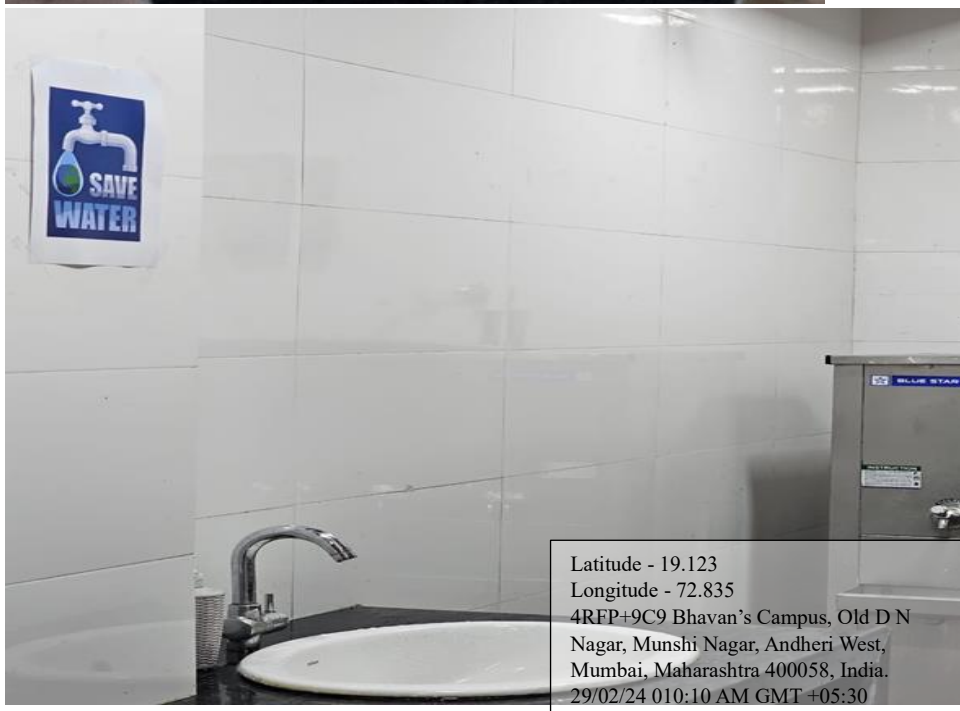
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Academic Year: 2022-23

WATER CONSERVATION MEASURES ON CAMPUS



Shraddha



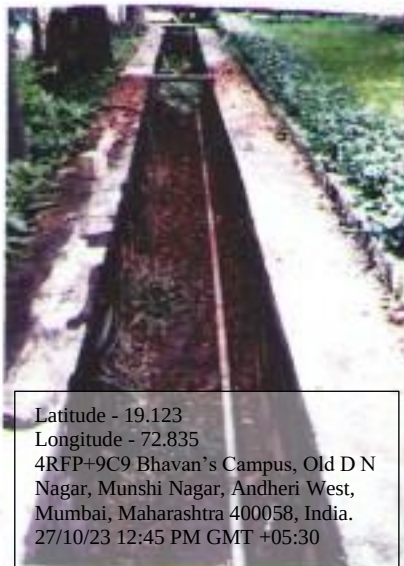


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STORMWATER DRAIN : RAINWATER DRAINS INTO THE LAKE



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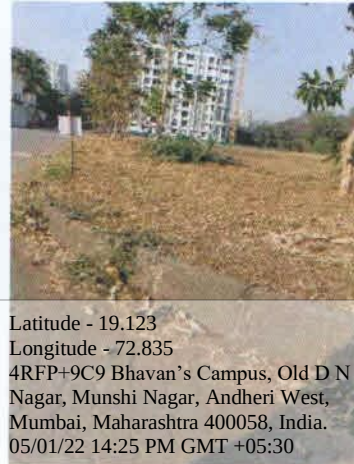
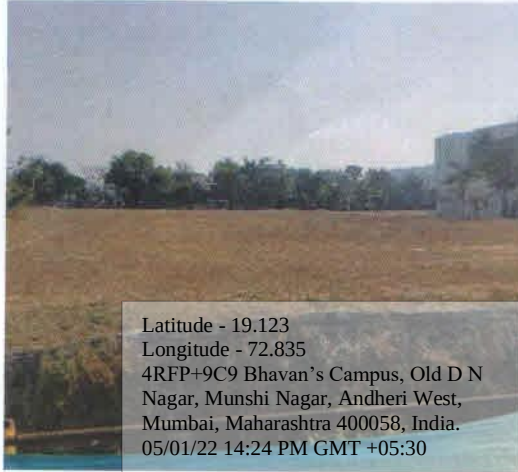


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Samir



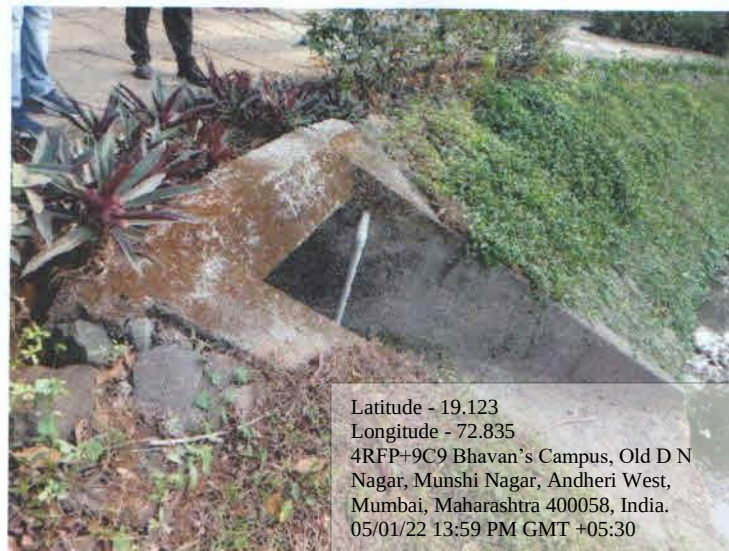
Bunds to prevent surface runoffs



Pipe carrying rainwater for gardening purpose



Outlet of Rain water Pipe to the lake



Schmidt
5/1/22

